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KARTHIKEYAN MUNIRATHINAM

A passionate PhD candidate with over 7 years of experience in academic teaching and research. Strongly aware of the state-of-the-art of research laboratory practice. Able to work independently and as part of a multi-disciplinary research and academic team. Demonstrated strong research ownership by conducting experiments on triboelectric nanogenerators for environmental energy harvesting and extending this study to IoT applications reflects both initiatives and practical perceptions. Experienced in programming specialized software to build real-time, self-sustainable, wireless electronic systems using nanogenerators, provides a potential benchmark for building next-generation technologies.

☐ PhD/Research guide

- **Prof. Dong-Weon Lee** [a, b, c]

  [a]MEMS and Nanotechnology Laboratory,

  [b]Advanced Medical Device Research center for cardiovascular disease

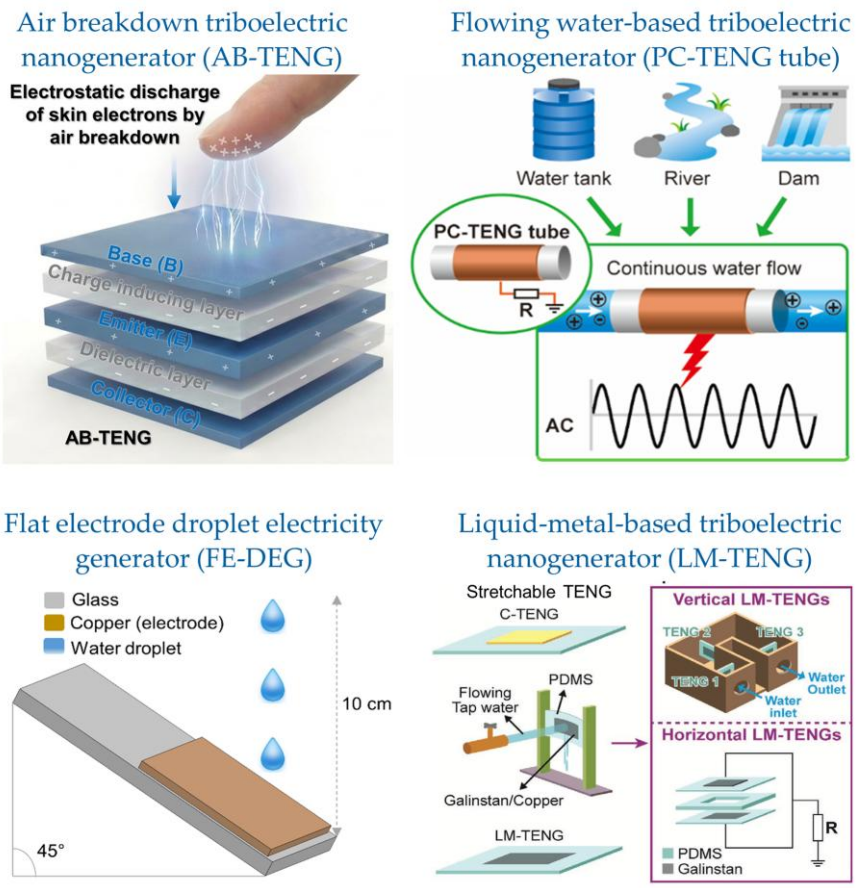
  [c]Center for Next Generation Research and Development, Department of Mechanical Engineering

  Chonnam National University, Gwangju, South Korea.

- e-mail: mems@jnu.ac.kr.

☐ Research Expertise

- **Triboelectric nanogenerators:** Proposed triboelectric nanogenerators (TENGs) for harvesting electric energy from the environment, such as human motion, skin interaction, water droplets, and flowing water.



- Currently working on triboelectric nanogenerators for harvesting electrostatic charges from the human body for self-sustainable, flexible, and wearable human-machine interface (HMI) systems.

☐ Research interest

- Triboelectric nanogenerators.
- Human-machine interfaces
- Real-time IoT systems.
- Wireless communication.
- Self-powered electronic systems
- Flexible nanogenerators
- Microfluidic devices.

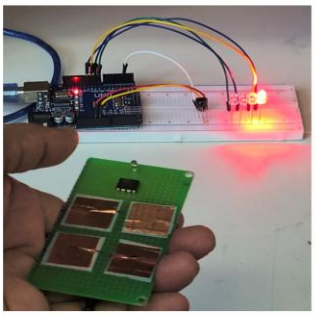
☐ Research statistics

☐ Research Expertise

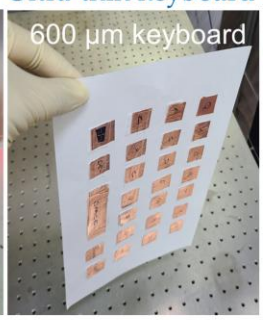
- **IoT modules:** Developed various real-world IoT modules by interfacing triboelectric nanogenerators (TENGs) with Arduino, LabVIEW, microcontrollers, Wi-Fi module, and web server.

AB-TENG-based human-machine interface (HMI) systems

IR remote control



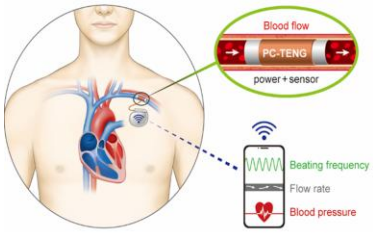
Ultra-thin keyboard



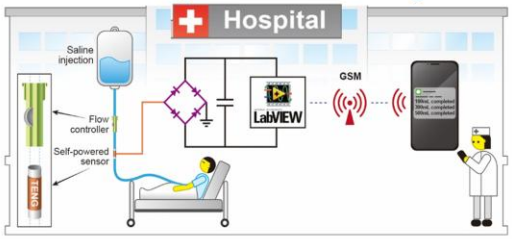
EG-TENG-based self-powered smart home

PC-TENG tube-based self-powered medical devices

Heart-rate monitoring



IV-drips level monitoring



- **Wireless communication:** Experienced in using various wireless protocols to demonstrate TENG-based IoT modules.
  - Infrared (IR) based wireless remote control and keyboard.
  - Wi-Fi-based web server for controlling smart home appliances through a mobile phone.
  - GSM-based IV-drips level indication through text/email from LabVIEW to mobile phone.
  - Inductive coupling for transmitting the resonant frequency of the wireless capacitive sensors.

- **Programming microcontrollers/IoT modules:** Experienced in writing source code for various IoT-based wireless protocols.
  - Programmed ATmega32u4, Atmel Tiny85, Arduino Pro Micro, and Arduino Uno to convert human signal into hexadecimal code for AB-TENG-based infrared (IR) remote control and keyboard.
  - Programmed Node MCU Wi-Fi module for controlling the smart home appliances through a web server.
  - Utilized LabVIEW and C# programs to send text/ email message through web platform for indicating the level of IV-drips.

Proficient skills

- **Software:** LabVIEW, Arduino, IoT modules, Solid Works, Origin, Auto CAD, C, C++, .NET, SQL Server, and HTML.
- **MEMS lithography process:** Mask aligner, spin coater, photo resists, hard & soft baking, liquid metal (Galinstan), PDMS, scanning electron microscopy (SEM), contact angle measurements, syringe pump, and optical microscope.
- **Measuring instruments:** Oscilloscope, source meter network analyzer, voltage & current pre-amplifier, and NI interface modules (DAQ, and FPGA).

Work Experience

February 2024 - February 2025, September 2025 – Present (1 year 6 months)

Post-doctoral Researcher, MEMS and Nanotechnology Laboratory, Chonnam National University, South Korea.

- Involved in developing triboelectric nanogenerators for harvesting electric energy from human-skin interactions.
- Involved in manuscript activities, such as scientific writing, drawing 3D diagrams, plotting graphs, and preparing supporting documents for the submission of the research article to the journal.
- Currently working on integrating triboelectric nanogenerators with AI for virtual-reality human-machine interfaces.

July 2014 – April 2015 (9 months)

Assistant Professor, Global Engineering College, Tamil Nadu, India.

- Involved in teaching for EEE, ECE, and CSE students.

August 2009 – May 2014 (4 years 9 months)

Assistant Professor, Sengunthar Engineering College, Tamil Nadu, India.

- **Subjects handled:** Microprocessor & Microcontrollers, Control Systems, Circuit Theory, Analog & Digital Electronics.
- **Labs handled:** Microprocessor & Microcontrollers Laboratory, Control Systems Laboratory, Analog & Digital Electronics Laboratory.

Education

Ph. D. from MEMS and Nanotechnology Laboratory- 2018 to 2024

Department of Mechanical Engineering, Chonnam National University, Gwangju, South Korea.

- Dissertation in "Flowing Liquid-Based Nanogenerators for Green Energy Harvesting and Self-Powered Sensor Systems".

M. Tech. in Sensor System Technology- 2007 to 2009

School of Electronics Engineering, VIT University, Vellore, Tamil Nadu, India.

- Coursework in "Design of Fuzzy Logic Controller and Real Time Implementation of PI Controller Using FPGA".

B. E. in Electrical and Electronics Engineering- 2001 to 2005

Shri Andal Alagar College of Engineering, Anna University, Chennai, Tamil Nadu, India

- Coursework in "Electrical $ Electronics Engineering".

Social media

- **Software:** LabVIEW, Arduino, IoT modules, Solid Works, Origin, Auto CAD, C, C++, .NET, SQL server, and HTML.
- **MEMS lithography process:** Mask aligner, spin coater, photo resists, hard & soft baking, liquid metal (Galinstan), PDMS, scanning electron microscopy (SEM), contact angle measurements, syringe pump, and optical microscope.
- **Measuring instruments:** Oscilloscope, source meter network analyzer, voltage & current pre-amplifier, and NI interface modules (DAQ, and FPGA).

Academic Achievements

Publications

International peer-reviewed journals (SCI)

- **Karthikeyan Munirathinam**, Longlong Li, Arunkumar Shanmugasundaram, Jongsung Park, Dong-Weon Lee, Air-Breakdown Triboelectric Nanogenertor Inspired by Transistor Architecture for Low-Force Human-Machine Interfaces. Nano-Micro Letters, 18, 2026 (IF: 36.3, JCR: 1%).
- **Karthikeyan Munirathinam**, Arunkumar Shanmugasundaram, Yun-Jin Jeong, Jong-Yun Kim, Dong-Weon Lee, High-Performance Triboelectric Nanogenerator Powered by Flowing Water for Self-Powered Wireless Sensor Platform. Chemical Engineering Journal, 505, 2025 (IF: 13.4, JCR: 3.3%).
- **Karthikeyan Munirathinam**, Kyeongha Kwon, Jongsung Park, Dong-Weon Lee, Liquid Dielectric Layer-Based Microfluidic Capacitive Sensor for Wireless Pressure Monitoring, Sensors and Actuators A: Physical, 357, 2023 (IF: 4.1, JCR: 20.4%).
- **Karthikeyan Munirathinam**, Gajula Prasad, Dong-Su Kim, Jongsung Park, Dong-Weon Lee, Liquid-Metal-Based Stretchable Triboelectric Nanogenerators for Flowing-Liquid-Based Energy Harvesting and Self-Powered Sensor Applications. Advanced Materials Technologies, 2201902, 2023 (IF: 6.4, JCR: 22.7%).
- **Karthikeyan Munirathinam**, Dong-Su Kim, Arunkumar Shanmugasundaram, Jongsung Park, Yun-Jin Jeong, Dong-Weon Lee, Flowing Water-Based Tubular Triboelectric Nanogenerators for Sustainable Green Energy Harvesting. Nano Energy, 102, 107675, 2022 (IF: 16.8, JCR: 5.3%).
- **Karthikeyan Munirathinam**, Jongsung Park, Yun-Jin Jeong, Dong-Weon Lee, Galinstan-Based Flexible Microfluidic Device for Wireless Human-Sensor Applications. Sensors and Actuators A: Physical, 315, 112344, 2020 (IF: 4.1, JCR: 20.4%).

International peer-reviewed journals (non-SCI)

- **Karthikeyan Munirathinam**, Arunkumar Shanmugasundaram, Dong-Weon Lee, Water-Based Triboelectric Nanogenertors: A Review. Micro and Nano Systems Letters, 13, 2025 (IF: 4.7, JCR: 38.7%).
- Arunkumar Shanmugasundaram, **Karthikeyan Munirathinam**, Dong-Weon Lee, SnO2 Nanostructure-Based Acetone Sensors for Breath Analysis, Micro and Nano Systems Letters 12 (3), 2024 (IF: 4.7, JCR: 38.7%).

Proficient skills

- **Software:** LabVIEW, Arduino, IoT modules, Solid Works, Origin, Auto CAD, C, C++, .NET, SQL server, and HTML.
- **MEMS lithography process:** Mask aligner, spin coater, photo resists, hard & soft baking, liquid metal (Galinstan), PDMS, scanning electron microscopy (SEM), contact angle measurements, syringe pump, and optical microscope.
- **Measuring instruments:** Oscilloscope, source meter network analyzer, voltage & current pre-amplifier, and NI interface modules (DAQ, and FPGA).

Academic Achievements

Conference proceedings

- **Karthikeyan Munirathinam**, Jongsung Park, Dong-Weon Lee, Microfluidic channel based stretchable wireless pressure sensor. Proceedings of the Korean Society of Precision Engineering Conference, 445, 2018.
- **Karthikeyan Munirathinam**, Jongsung Park, Dong-Weon Lee, Liquid-metal-based flexible microfluidic device for wireless sensor applications. International Conference on Optical MEMS and Nanophotonics (OMN), 68-69, 2019.
- **Karthikeyan Munirathinam**, Gajula Prasad, Dong-Weon Lee, Fabrication of micro-hole-patterned tubular triboelectric nanogenerator for self-powered ocean-monitoring systems. Proceedings of the Korean Society of Precision Engineering Conference, 400, 2022.

Conference attended

- **International conference: 3, Domestic conference: 13**

Research Goal

To utilize my experience and subject/programming knowledge to interface triboelectric nanogenerators with artificial intelligence for building real-world smart IoT modules for various virtual-reality applications.

Declaration

Hereby, I declare that the information provided above is true to the best of my knowledge.

Sincerely

KARTHIKEYAN MUNIRATHINAM